# Depth-Width Tradeoffs for Transformers on Graph Tasks

*Linear width can replace logarithmic depth — for some tasks*

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**⋆ TL;DR**

Sanford et al. 2024a sorted graph tasks by the **depth** a narrow transformer needs (connectivity: $\Theta(\log N)$ layers at width $N^{\varepsilon}$). This paper asks the opposite question: **fix the depth and let the width grow — how wide must it be?** With each node's **adjacency row** as its token, they build a **width hierarchy** (Figure 1): node degree needs $O(\log n)$ width; 1-vs-2 cycles and $A^L$ need $O(n)$; counting a $k$-node subgraph needs at most $O\left(n^{2-\frac{1}{k}}\right)$; Eulerian-cycle verification needs nearly $n^2$; and $n^2$ suffices for **anything**, because the whole graph then fits in one token. Upper bounds are explicit constructions that lean on **arbitrarily powerful MLPs**; lower bounds come from communication complexity (unconditional for 2-cycle detection, conditional on the 1-vs-2-cycle conjecture elsewhere). The experiments show that at a fixed parameter budget, shallow-wide models train and infer faster with the same accuracy, and that the width needed to **fit** 4-cycle counts grows roughly linearly in $n$.

**✦ The big question**

"What are the algorithmic capabilities obtained by increasing the width of a transformer?" — measured as the width needed, at constant depth, as a function of the number of nodes $n$.

**✦ The one-line answer**

It depends on the task: some need only $O(n)$ width at constant depth, some need super-linear width, one needs nearly $n^2$. Width is a real substitute for depth — but only up to the information in the input.

## 1 · Setup

### The transformer they analyse (§3.1)

**▸ Definition — Model**

Input tokens $x_1, \ldots, x_N \in \mathbb{R}^{d_{\text{in}}}$. Each layer: multi-head self-attention with $H$ heads, then a residual connection, then a per-token MLP. **Each head has its own $m \times m$ key, query and value matrices** ($K_h, Q_h, V_h \in \mathbb{R}^{m \times m}$), and the head outputs are summed. Width $m$ = embedding dimension. Bit precision $p = O(\log n)$.

Three assumptions do most of the work in the proofs:

1. **The MLP is arbitrary.** It may compute any function of a single token (and of its position). This is standard in this line of theory (Sanford et al. 2024b) and isolates what **attention** can and cannot move between tokens. The MLPs in their constructions are polynomial in $n$, but some are huge (Remark 5.3: exponential in $k$ for subgraph counting).
2. **Positional information is free.** The MLP sees the token index $i$, so a token knows "I am node $i$".
3. **Logarithmic precision.** Every number carries $O(\log n)$ bits. The lower bounds count bits, so this assumption decides how strong they are (see §4).

**✎ My note**

**Mapping to our model.** Their $H$ heads each have the full width $m$, so a layer has $mH$ "channels". Ours split the width: head size 8, $H = m/8$, so all heads together span exactly $m$. Our $m$ therefore plays the role

of their $m$ (one head), and of their $mH$ in Sanford's bounds. Our MLP is $4m$ wide with ReLU — far from arbitrary — and we train in 32-bit floats, not $O(\log n)$ bits.

### The tokenization: adjacency rows (§3.1, "Graph inputs")

▸ **Definition — Node-adjacency tokenization**

One token per node: token $i$ is row $i$ of the adjacency matrix $A \in \{0,1\}^{n\times n}$. So $N = n$ tokens, each $n$ numbers wide.

**Why not Sanford's edge list?** One token per edge gives $N = n + |E|$ tokens; for a dense graph that is $\Theta(n^2)$ tokens, and attention's quadratic cost becomes $\Omega(n^4)$. Node-wise tokens keep the sequence at $n$. Like the edge list, adjacency rows are **not** permutation invariant (a token's coordinates are tied to node numbering).

**My note**

This is **our** tokenization (we add self-loops: $A + I$). So their theorems are the relevant theory for our models, not Sanford's — Sanford's $N$ and $\sqrt{N}$ are stated for edge tokens. Consequence: an adjacency-row token already carries $n$ bits (its whole neighbourhood). Every bound below has to be read with that in mind: a lot of the "work" is done by the input before any attention happens.

## 2 · The hierarchy at a glance (Figure 1)

| Task | Width that suffices | Width that is too small | Source |
|---|---|---|---|
| node degree | $O(\log n)$ (even $m = 1$, App. A) | — | Sanford 2024a; App. A |
| 2-cycle detection, max degree $d$ | $O(d \log n)$, 1 layer | $mpHL = o(d)$ | Thm 4.4, B.3 |
| 2-cycle detection (directed) | $O(n)$, 2 layers | $mpHL = o(n)$ — **unconditional** | Thm 4.3, 4.2 |
| $A^L$ (walk counts) | $O(n)$, $O(L)$ layers | — | Thm 4.3 |
| 1 vs 2 cycles | $O(n)$, 2 layers | $O(n^{1-\varepsilon})$ at depth $o(\log n)$ — conditional | Thm 4.1; Sanford 2024a |
| connectivity (general) | $O(n \log^3 n)$, constant depth — **sketch only**, randomized | depth $\Omega(\log n)$ or width $\Omega(n)$ (App. A) | §4.1, Ahn et al. 2012 |
| count / detect a $k$-node subgraph | $O\left(n^{2-\frac{1}{k}}\right)$, $O(1)$ layers | — | Thm 5.2 |
| Eulerian cycle verification | $n^2$ | $O(n^{2-\varepsilon})$ at depth $o(\log n)$ — conditional | Thm 5.1 |
| any function $f(A)$ | $O(n^2)$: the whole graph fits in one token | — | §5 |

**My note**

Read the table as "width needed at **constant** depth". The two conditional lower bounds rest on the **1 vs 2 cycles conjecture** (App. B.1): an MPC protocol with $n^{1-\varepsilon}$ memory per machine cannot tell one $n$-cycle from two $n/2$-cycles in $o(\log n)$ rounds. It is widely believed, unproven, and the backbone of Sanford et al. 2024a's depth lower bounds too.

# 3 · The positive results — how width replaces depth

## Theorem 4.1 — one vs two cycles with two layers

◆ **Theorem 4.1**

There exists a transformer with 2 layers of self-attention and embedding dimension $O(n)$ that solves the 1 vs 2 cycle problem.

**How (App. B.2).** The trick is to put the **whole graph into one token** and let the MLP solve it.

1. Layer 1: attention does nothing ($V = 0$, the residual keeps the input). The MLP turns node $i$'s adjacency row into the IDs of its two neighbours $(j, k)$ — possible because $j + k$ and $j^2 + k^2$ determine $\{j, k\}$ (Lemma B.2) — and writes $(i, j, k)$ into its own 3 slots of a $3n$-dimensional vector.
2. Layer 2: attention with $K = Q = 0$ is uniform, i.e. every token receives the **average** of all tokens; $V = nI$ turns it into the **sum**. Now every token holds the full edge list ($3n$ numbers).
3. The MLP runs any connectivity algorithm on that edge list (universal approximation).

✎ **My note**

This is less a "transformer algorithm" than a **compression argument**: a cycle has only $n$ edges, so the graph fits in $O(n)$ numbers. The authors say it themselves — the result "demonstrates the brittleness of communication-based transformer lower bounds, since even a slight increase in the embedding dimension breaks them", and "is not surprising, since the graph in this task is very sparse". It needs one layer of **attention** that is uniform — no routing along paths at all.

### General connectivity: only a sketch

A dense graph has up to $n^2/2$ edges, which do not fit in $O(n)$ numbers. For general connectivity they **sketch** (not a theorem): compress each adjacency row with the linear sketches of Ahn, Guha and McGregor (2012) to $\mathrm{poly}(\log n)$ numbers, gather all sketches into one token ($O(n \log^3 n)$ total), and let the MLP run Ahn et al.'s algorithm. It works only with high probability (random projections), "thus, we don't present it here formally".

! **Check against our thesis**

Our thesis says that with adjacency rows, constant depth and $O(n)$ width suffice for connectivity (abstract l. 17–19; ch. 1 summary; ch. 3 §Хэлэлцүүлэг "бүтээц $O(n)$ өргөн ашигладаг"). **Strictly:** the formal constant-depth, $O(n)$-width theorem is for 1-vs-2 cycles (and 2-cycles); for general connectivity it is a randomized sketch with $O(n \log^3 n)$ width. Appendix A states the adjacency tokenization needs "either depth $\Omega(\log n)$ or width $\Omega(n)$" for connectivity.

## Theorem 4.3 — powers of $A$, one hop per layer

◆ **Theorem 4.3**

There exists an $O(L)$-layer transformer with embedding dimension $m = O(n)$ such that, for any graph embedded as rows of $A$, the output at token $i$ is row $i$ of $A^L$.

**How (App. B.4).** Each token carries $\left(A_i, e_i, A_i^\ell\right)$ — its adjacency row, a one-hot position, and the current power. Keys and queries are chosen so that the attention score between $i$ and $j$ is $c \cdot A_{ij}$; with a large temperature $c$ the softmax becomes a hard **average over $i$'s neighbours**. The value vectors carry $A^\ell$, so the attention output is $A \cdot A^\ell$ divided by the degree; the MLP multiplies the degree back in. Each layer multiplies by $A$ **once**: $A^\ell \to A^{\ell+1}$.

! **Check against our thesis**

Our ch. 1 (ch1-theory.tex, "Гүн ба өргөний солилцоо", first bullet) says this construction is "the transformer form of repeated squaring" ($M \to M^2$). **It is not:** it multiplies by $A$ once per layer, so $L$ layers give $L$ hops.

Repeated squaring would give $2^L$ hops; deciding connectivity through $A^L$ this way needs $L \approx$ diameter (the paper says "checking whether $A^n$ doesn't contain any zero entries" — $n$ layers).

💡 **Thesis hook**

This is the "one hop per layer" hypothesis our probes tested, and they found neither it nor doubling: 3-hop information appears after layer 1 and then stays flat, while the "connected" signal grows a little in every layer (progress report, Probing). Our 6-layer models solve graphs of diameter 10–13, so they cannot be running Theorem 4.3's algorithm.

### Theorem 4.4 — sparse graphs need much less width

◆ **Theorem 4.4**

For any $n$ and $d \le n$, there exists a single-layer transformer with embedding dimension $O(d \log n)$ that detects 2-cycles in any graph with maximum degree $d$. This is optimal up to log factors.

**How (App. B.5).** Give each node a random code $y_i \in \mathbb{R}^p$, $p = O(d \log n)$, and encode a neighbour set $x$ (at most $d$ ones) as $\varphi(x)$ with $\langle \varphi(x), y_i \rangle = 1$ if $i \in x$ and $\le 1/2$ otherwise (restricted isometry, Candès–Tao). Queries carry $(\varphi(x_i), y_i)$, keys $\big(y_j, \varphi(x_j)\big)$, so the score is large exactly when $j \in x_i$ **and** $i \in x_j$, i.e. when $i \to j \to i$ is a 2-cycle. A "dummy node" absorbs the attention when no 2-cycle exists.

✎ **My note**

Width scales with the **degree**, not with $n$: the adjacency row is mostly zeros and can be compressed. Our `swap` graphs have degree $\approx 2.5$, so by this logic a narrow model could represent local tasks. Our Q6 test (fixed random $n \times 96$ projection of the adjacency row) is the same idea — random node codes — and showed that the input compression is not what drives the growth of $m^*$.

### Theorem 5.2 — subgraph counting between linear and quadratic

◆ **Theorem 5.2**

For a fixed $k$-node pattern $G'$, there is a transformer with $O(1)$ layers and width $O\big(n^{2-\frac{1}{k}}\big)$ that counts the occurrences of $G'$ in any $n$-node graph.

**How (App. C.1).** The "tri, tri again" algorithm (Dolev et al. 2012). Split the nodes into $n^{1/k}$ groups of $n^{1-1/k}$. Layer 2 gathers each group's internal edges into one token ($\le n^{2-2/k}$ edges). Layer 3 lets each token collect one **combination of $k$ groups**: there are $\binom{n^{1/k}}{k} \le n$ combinations, so they fit in the $n$ tokens. Every copy of $G'$ lies inside some combination; the MLP counts there.

✎ **My note**

The MLP is exponential in $k$ (Remark 5.3), and some copies are counted in several combinations — fine for **detection**, needs care for exact counts. Compare Sanford et al. 2024a Thm 23: depth $O(\log \log n)$ but $O\big(n^{k-1}\big)$ blank tokens. Width and extra tokens are two currencies for the same work.

## 4 · The negative results — when width cannot be small

### Theorem 4.2 — 2-cycle detection needs linear width (unconditional)

◆ **Theorem 4.2**

Let $T$ have width $m$, depth $L$, precision $p$ and $H$ heads per layer, with adjacency-row inputs. If $T$ detects 2-cycles in directed graphs then $mpHL = \Omega(n)$ with residual connections, and $mpH = \Omega(n)$ without.

**How (App. B.3): set disjointness.** Alice holds $a \in \{0,1\}^{n^2}$, Bob holds $b$; they must decide whether some $a_i = b_i = 1$, which needs $n^2$ bits of communication. Build a graph on $2n$ nodes: $a$ = edges from side 1 to side 2, $b$ = edges back. A directed 2-cycle exists iff the sets intersect. Alice owns side 1's tokens, Bob side 2's. To simulate one layer, Bob sends Alice his keys, queries and values: $O(nmpH)$ bits. Over $L$ layers, $O(nmpHL) \geq n^2$, hence $mpHL = \Omega(n)$.

> 💡 **Thesis hook**
>
> **Plug in our numbers.** In our model the per-layer message is about $3m$ numbers per token (the heads split $m$), at $p = 32$ bits. The bound becomes $3 \cdot 32 \cdot m \cdot L \geq n$; at $L = 6$ that is $m \geq n/576$ — trivially satisfied. Lower bounds of this kind count **bits**, and a 32-bit float carries far more than the $O(\log n)$ bits the theory assumes. So **no communication bound explains our measured $m^* \approx 0.7n$–$1.0n$**: the binding constraint is what gradient descent can learn to use, not what the channel can carry. This sharpens the "representation vs learning" argument in our ch. 3.

### Theorem 5.1 — Eulerian cycles need nearly quadratic width (conditional)

> ◆ **Theorem 5.1**
>
> Under the 1 vs 2 cycles conjecture, Eulerian cycle verification on multigraphs (with adjacency inputs) cannot be solved with $m = O(n^{2-\varepsilon})$ unless $L = \Omega(\log n)$.

**How (App. C.2).** Reduce 1-vs-2 cycles on $N = n^2$ nodes to Eulerian verification on an $n$-node multigraph: map each original node to one of $n$ super-nodes, keep all $n^2$ edges, and encode the cycle's successor structure as path fragments. One cycle $\leftrightarrow$ the fragments form one Eulerian cycle. A cheap transformer does the conversion, so a sub-quadratic-width solver would break the conjecture.

> ✎ **My note**
>
> The only task shown to need (nearly) $n^2$ width — and it is contrived. It does establish that the trivial "copy the whole graph" upper bound can be tight.

## 5 · Appendix A — tokenization trade-offs

- **Adjacency rows are good at local tasks**: degree with $m = 1$ ($\langle \mathbf{1}, x_i \rangle$). Laplacian eigenvectors (the $m$ smallest) cannot even give degrees without $m$ growing with $n$ (footnote 6: $n/3$ disjoint paths make the eigenvectors component indicators).
- **Laplacian eigenvectors are good at global tasks**: the multiplicity of eigenvalue 0 is the number of components, so connectivity is **free** given the eigenvalues — while adjacency rows need "either depth $\Omega(\log n)$ or width $\Omega(n)$".
- **At $m = \Omega(n^2)$ both are universal** — one token holds everything.

> 💡 **Thesis hook**
>
> This is the theory behind our encoding chapter: an encoding that precomputes global structure can give the answer away. Our RWSE-16 leak and the $k$-hop rows are the same phenomenon in degrees: the more of the global computation the input already contains, the less width the model needs.

## 6 · Experiments — what they measured, and how it compares to ours

### §6.1 — same parameter count, different shape

- Tasks: connectivity (their own generator), triangle count and 4-cycle count (Chen et al. 2020, Erdős–Rényi $p = 0.1$). 5000 graphs each; $n = 50$ and 100.

- About 100k parameters split as $(L, m) \in \{(1, 125), (2, 89), (4, 63), (8, 45), (10, 40)\}$; 100 epochs; 3 seeds. Learning rate tuned in $\{10^{-4}, 5 \cdot 10^{-5}\}$, batch size in $\{32, 64\}$, dropout 0.1.
- Result: loss and accuracy are about the same for every shape; shallow-wide is much faster to train and run (GPUs parallelize within a layer, not across layers).

> **My note**
>
> **Their connectivity data is easy.** Classes come from mixtures of generators (ER, random geometric, Barabási–Albert, SBM) with connected graphs drawn at **higher density** than disconnected ones ($p \geq \ln n/n$ for connected ER, "a lower $p$" for disconnected). Our reproduction ( `docs/yehudai-empirical.md` ) found that a **1-layer, width-64** adjacency-row model reaches 100% test accuracy at $n = 50$ — with 100 training graphs in their own code, by epoch 4 in ours. So "every depth gives the same accuracy" says little about depth on connectivity: one layer already suffices on this data. (The density correlation is our reading of their generator description, not their claim.)

### §6.2 — the critical width

- **Definition:** the width at which the **training loss** plateaus above 0.05, i.e. the smallest width that can **fit** the training set. Grid search over hyperparameters, best fit kept.
- Setup: **one layer, two heads**; $n = 50$–$400$ (step 50), $m = 100$–$800$ (step 100); triangle and 4-cycle counting.
- Result: "the critical width increases roughly linearly with the graph size" (Fig. 3; triangles in Fig. 9). The text also says "sub-quadratically … as argued in Section 5.2".

> **My note**
>
> Four differences from our measurement, each of which we found matters:
>
> 1. **Fit, not generalization.** On 5000 graphs a wide model can memorize; our Q1c and the 32k-graph fine grid show the two critical widths separating when data is scarce.
> 2. **A two-value learning-rate grid** ($10^{-4}$, $5 \cdot 10^{-5}$), with widths from 100 to 800: our Q1 shows the best rate moves with width, by more than a factor of two over such a range.
> 3. **Coarse width grid** (steps of 100), so each critical width is known to within ±50.
> 4. **A local task** (counting small patterns), at depth 1. Connectivity is global.
>
> Their "linear" and our $n^{1.74}$ are therefore not in conflict — different task, depth and criterion — but theirs is the only prior critical-width measurement for graph transformers, and ours is its stricter extension.

### §6.3 — tokenizations on real molecules

ROC-AUC on three OGB molecule datasets (3 seeds):

| Tokenization | MOLHIV | MOLBBBP | MOLBACE |
|---|---|---|---|
| edge list | 54.0 ± 1.4 | 64.7 ± 1.7 | 66.1 ± 3.9 |
| adjacency rows | 61.9 ± 1.1 | **67.6 ± 2.6** | **68.6 ± 2.3** |
| Laplacian eigenvectors | **68.1 ± 1.5** | 55.3 ± 4.8 | 63.6 ± 2.3 |

Adjacency rows beat the edge list everywhere and the Laplacian on two of three. Their argument for the tokenization: it works on real data, not only in proofs.

## 7 · Checking our thesis against the paper

| Where | What we wrote | Verdict |
|---|---|---|
| bibliography | title "…in Algorithmic Reasoning of Graph Tasks with Transformers", ICML 2025 | **wrong**: published title "Depth-Width Tradeoffs for Transformers on Graph Tasks", NeurIPS 2025 (the arXiv listing keeps the old title) |

| | | |
|---|---|---|
| ch. 1, “Гүн ба өргөний солилцоо” | $O(L)$ layers, $O(n)$ width compute $A^L$ — “the transformer form of repeated squaring” | **wrong**: one hop per layer (§3 above) |
| ch. 1, same | 1 vs 2 cycles: 2 layers, $O(n)$ width; sub-linear width at constant depth fails only conditionally | correct (Thm 4.1, §4.1) |
| ch. 1, same | critical width for 4-cycle counting grows linearly in $n$ | correct; add: **training-loss** criterion, 1 layer |
| ch. 1, “Өргөнийг туршилтаар…” | they used **only one** tokenization and kept the learning rate **fixed** across widths | **imprecise**: they compare three tokenizations on OGB (§6.3), though the critical-width study uses only adjacency rows; the rate was tuned, but only over $\{10^{-4}, 5 \cdot 10^{-5}\}$ |
| ch. 2, §Сургалтын хурд | fixing the rate across widths is common, citing them | same nuance: two-value grid |
| abstract; ch. 1 summary; ch. 3 discussion | with adjacency rows, constant depth and $O(n)$ width suffice (for connectivity) | **overstated**: formal only for 1-vs-2 cycles / 2-cycles; general connectivity is a randomized $O(n \log^3 n)$ sketch |
| ch. 1, Merrill comparison | their construction gives each token $n$ bits and allows arbitrary MLPs | correct (§3.1) |
| progress report, conclusions | critical width “measures **fitting** on counting tasks at depth 1 with 5000 graphs and a fixed learning-rate grid” | correct |
| progress report, conclusions | “quadratic width suffices for any task by copying the whole graph” | correct (§5) |

## 8 · Hooks into our work

**💡 Thesis hook**

- **Representation vs learning, made concrete.** Their theory gives $O(n)$ (and Thm 4.2's $mpHL = \Omega(n)$, which is trivially met in floating point). We measure the width gradient descent needs: $n^{1.74}$ at depth 6, $\approx n^3$ at depth 2 (Q7c). Their §7 lists “optimization and generalization” as future work — that is exactly our gap.
- **Depth 2 is their regime.** Thm 4.1 is a 2-layer, $O(n)$-width construction. Our Q7a found 2 layers learn connectivity at $m \approx 1.1n$ at $n = 40$ — in line — but Q7c found the 2-layer critical width growing $\approx n^3$ across sizes. Gradient descent does not find the “gather everything into one token” solution as $n$ grows.
- **Their mechanism vs ours.** Theorems 4.1 and 4.3 imply two concrete internal pictures: “the whole graph in one token” and “one more hop per layer”. Our probes rule out the second and find no per-node reachable set (the first would show up as nodes storing global information). The model compares pairs of vectors instead.

**? Open question**

Does the $n^{2-\frac{1}{k}}$ bound for counting predict a **super-linear** fitted critical width at larger $n$ that their coarse grid could not resolve? Their Fig. 3 points span $n = 50$–$400$ with width steps of 100.

## 9 · One-paragraph summary

Yehudai et al. turn Sanford et al.'s depth question around: at **constant** depth, how wide must a transformer be for a graph task, when each node's adjacency row is its token? Their hierarchy runs from $O(\log n)$ (degrees) through $O(n)$ (1 vs 2 cycles, 2-cycles, $A^L$ with $O(L)$ layers) and $O\left(n^{2-\frac{1}{k}}\right)$ (subgraph counting) to nearly $n^2$ (Eulerian cycle verification), with $n^2$ always enough. The constructions rely on arbitrary MLPs and often on packing the graph into a single token; the lower bounds count bits through the attention layer, unconditionally for 2-cycles and conditionally elsewhere. For general connectivity there is only a randomized sketch. Experimentally, shallow-wide models match deep-narrow ones at equal parameters and run faster, and the width needed to fit small-pattern counts grows about linearly. For our thesis it is the theory of our exact tokenization, the source of the "critical width" method, and the baseline our connectivity measurements ($n^{1.74}$, learned, generalizing) extend and qualify.